Werner **Bisschoff**

Hybrid Edge/Systems Engineer

AWS Certified Solutions Architect – Associate (In Progress — Target: 6 Weeks)

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Hybrid Edge/Systems Engineer with 5+ years of experience architecting end-to-end telemetry systems spanning edge compute to backend integration. Specializes in C/C++ and FreeRTOS for deterministic real-time scheduling, Python for automation and hardware-in-the-loop validation, and Elixir/OTP for distributed platform infrastructure.

Skills

Primary Competencies : C, C++, FreeRTOS, ESP32/ESP-IDF, NFC APDU (ISO 14443-4)
Secondary Competencies : NimBLE, Bare-Metal Runtimes, UART/SPI/I2C, Hardware-in-the-Loop (HIL) Testing
Foundational Systems : Linux IPC, Bash, Python (pytest, Custom Automation Frameworks), Assembly, System Modeling
Cross-Domain Integration : ERPNext (Integration-focused), Docker, Spec-Driven Development (SDD)

Experience

Principal Systems Architect — Cape Town

Divergent Tabletop — Jul 2025

- Engineered a distributed real-time data engine utilizing Elixir/OTP and conflict-free replicated data types (CRDT models). Deployed in-memory GenServer storage buffers with threshold-based flush mechanics, reducing direct database write transactions by approximately 85% under sustained edit workloads.
- Integrated a Rust-backed CRDT synchronization layer (y_ex) using actor-based concurrency, achieving sub-50ms convergence latency across distributed editing sessions through deterministic state reconciliation and binary delta serialization.
- Designed a dynamic BEAM supervision tree with one_for_one isolation, encapsulating each page editing session in its own process subtree to prevent cascading failures and guarantee per-session state integrity.

Embedded Systems & Integration Engineer — Cape Town

FARO Africa — Aug 2024 – Nov 2025

- Designed ISO 14443-4 NFC communication architectures for e-paper display nodes and secure card reader matrices.
- Engineered lock-free asynchronous device management services bridging hardware peripherals with application layers.
- Built Python-based hardware peripheral simulators to catch edge-case device state corruptions prior to flashing.
- Introduced LLM-assisted development workflows to improve debugging speed and code review throughput.

Firmware Engineer (Contract) — Remote

Ingenics Digital GmbH (through ViVa Outsourcing) — Mar 2023 – May 2024

- Architected deterministic FSMs on dual-core ESP32 using FreeRTOS for concurrent sensor arrays.
- Developed Cap'n Proto IPC over serial lines, achieving sub-millisecond serialization latency.

Software Developer — Cape Town

UMAN Technologies — Mar 2021 – Dec 2022

- Used perf profiling and flamegraphs to trace a performance bottleneck to JSON serialization and verbose logging in a SOME/IP request handling loop. Eliminated unnecessary logging via config toggle reducing latency by approximately 80%, then drove an architecture redesign to phase out JSON entirely.
- Sole owner of an IPC service layer — studied existing node schema, designed node structures and data contracts, and implemented the communication layer enabling the frontend to consume the service.
- Built Python-based SOME/IP hardware mocks simulating request/response behavior; integrated into CI so automated tests ran without physical hardware.

Education

B.Eng. Computer and Electronic Engineering — Potchefstroom

North-West University — 2020

Focus on embedded systems, software engineering, and electronic design · Developed NFC payment protocol emulator (Kotlin/Android), STM32 firmware (C), PID motor controller (Arduino)